MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE GDB

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389 except as modified in 3.4.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.05 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389 except the operating temperature range shall be 0° to 80°C.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GDB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/10-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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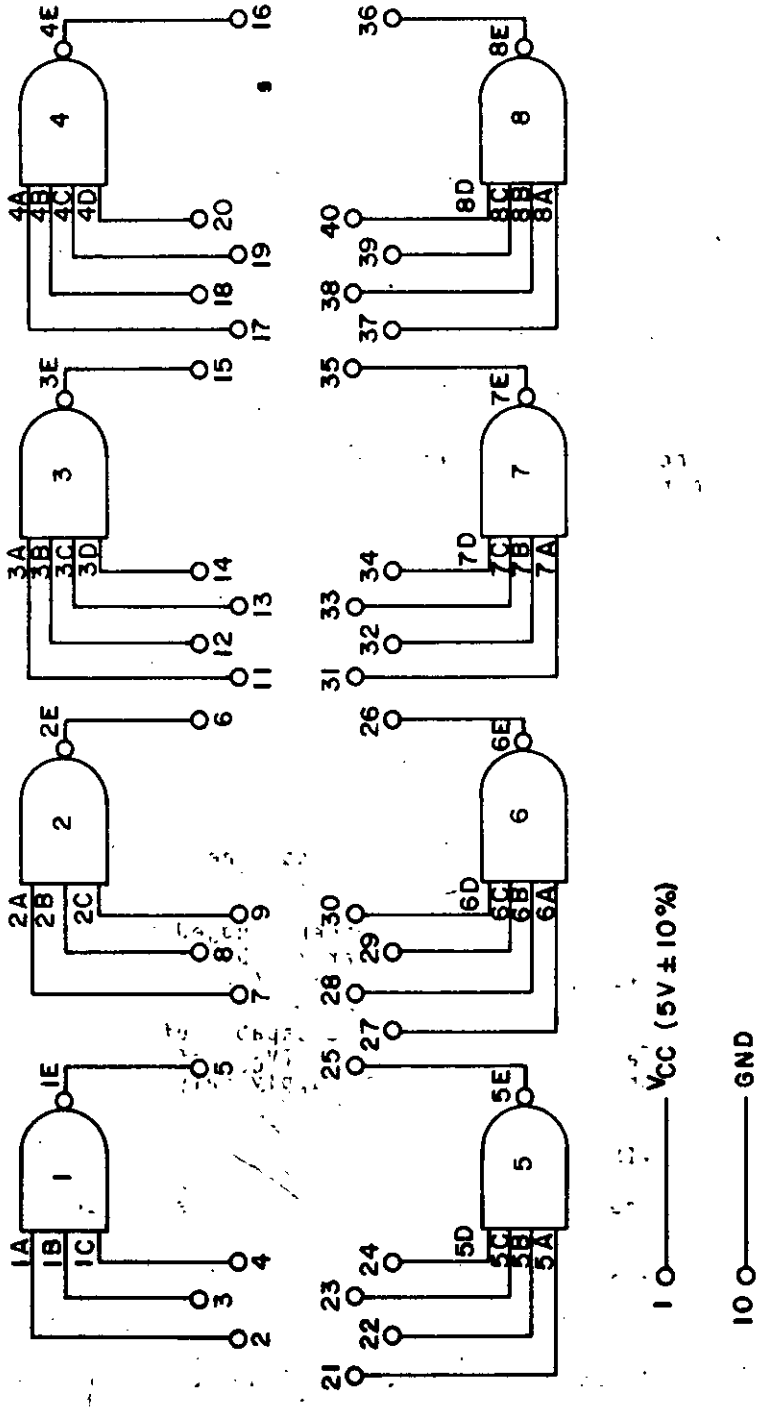


FIGURE 1. GDB module function diagram, NAND gate.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 80°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A8	V_{OUT} (low-loaded)	III	S	0.1	0.8	0.1	0.8	0.1	0.8	0.1	0.8	V
B1-B30	Input current	IV	S	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	mA
C1-C30	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D8	Output voltage, low level	VI	X	0	450	0	450	0	450	0	450	mV
E1-E30	Output voltage, high level	VI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F30	Input current, low level	VII	X	1.0	2.2	1.0	2.2	1.0	2.2	1.0	2.2	mA
G1-G84	Input current, high level	VII	X	0	60	0	60	0	60	0	60	µA
H1-H8	Output short circuit current	VII	X	40	100	35	100	35	100	25	100	mA
J1	Power supply current, low level	VIII	S	20	90	20	90	20	90	20	90	mA
J2	Power supply current, high level	VIII	S	10	44	10	44	10	44	10	44	mA
K1-K8	Transition time (t_{THL})	IX	S	3	20	3	20	3	25	3	25	ns
L1-L8	Transition time (t_{TLH})	IX	S	3	20	3	20	3	25	3	25	ns
M1-M8	Propagation delay time (t_{PHL})	X	S	3	20	3	20	3	25	3	25	ns
N1-N8	Propagation delay time (t_{PLH})	X	S	3	20	3	20	3	25	3	25	ns
P	Power supply transient amplitude	XI	Q	0	400	0	400	0	400	0	400	mV
Q	Isolation	XII	S	10	∞	10	∞	10	∞	10	∞	MΩ
R1-R8	Output crosstalk, low level	XIII	Q	0	400	0	400	0	400	0	400	mV

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 80°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S1-S8	Output crosstalk, high level	XIII	Q	0	400	0	400	0	400	0	400	mV
T1-T30	Input capacitance	XIV	Q	2	12	2	12	2	12	2	12	pF
U	Power supply transient rise time	XV	S	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	µs

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through N, R, S, and U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through U shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/10 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No 10001-2658754. Part number M28787/10-1 supersedes part numbers 2658754 and M28787/10.

* 6.4 Module description. The following electrical parameters reflect the 0° to 80°C end-of-life limits from table I.

Supply voltage	5.0 Vdc ±10%
Maximum power dissipation	
All outputs at low level	495 mW (outputs not loaded)
All outputs at high level	242 mW (outputs not loaded)

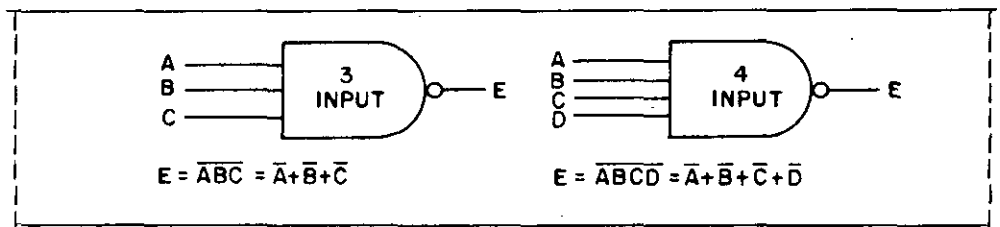
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Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, 5.5 V maximum
Input current	
Low level (V_{IL} = 0.V)	2.2 mA maximum
High level (V_{IH} = 2.4 V)	60 µA maximum
Output voltage	
Low level	0.0 V minimum, 0.45 V maximum
High level	2.4 V minimum, V_{CC} maximum
Output drive current	
Low level (V_{OL} = 0.V)	22 mA maximum
High level (V_{OH} = 2.4 V)	0.6 mA maximum
Propagation delay time	
t_{PLH}	25 ns maximum
t_{PHL}	25 ns maximum

* 6.5 Module function. This module contains six four-input and two three-input NAND gates.

6.5.1 Input to output relationship. Each gate has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.



Input	Output
Any inputs at low level	High level
All inputs at high level	Low level

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

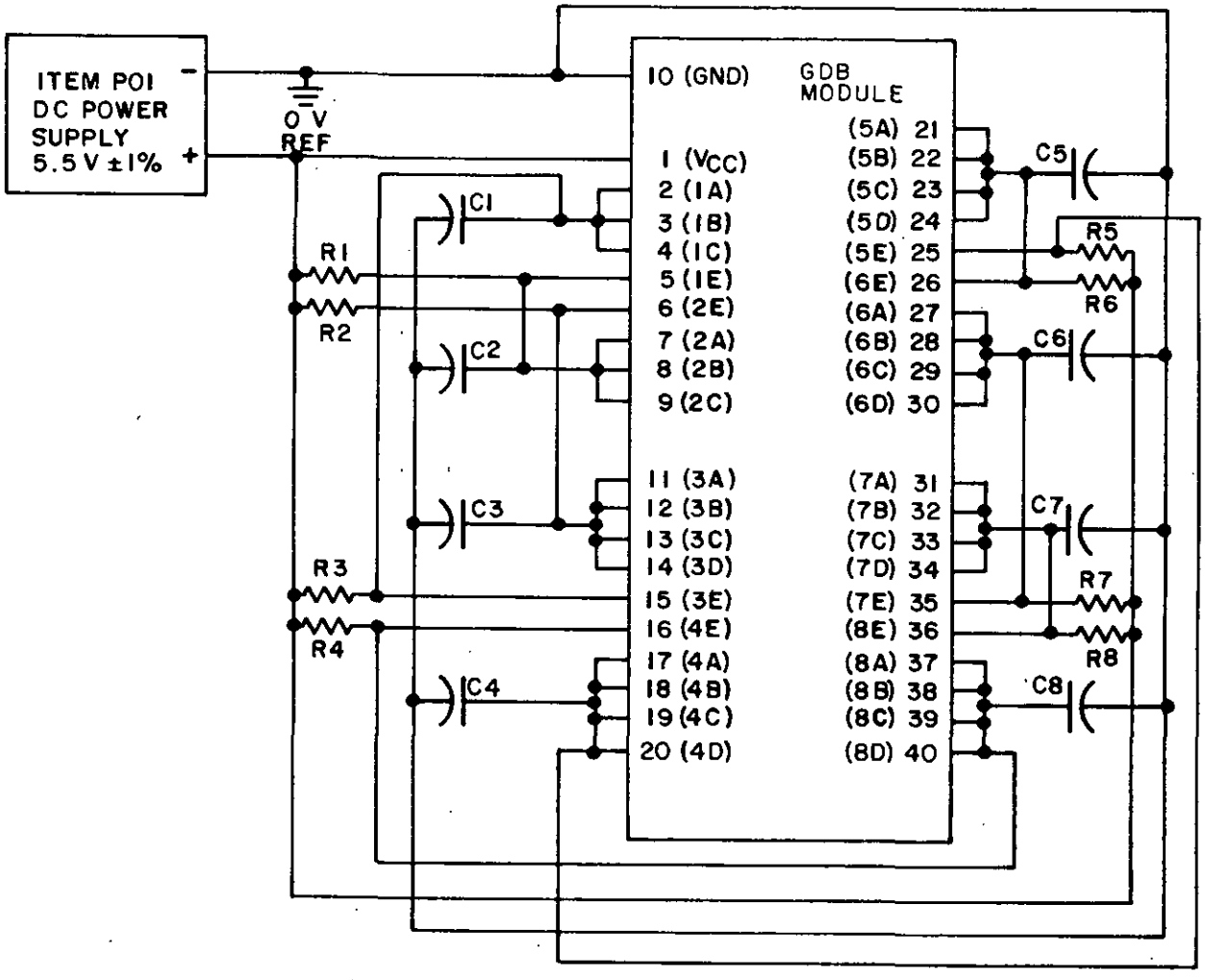
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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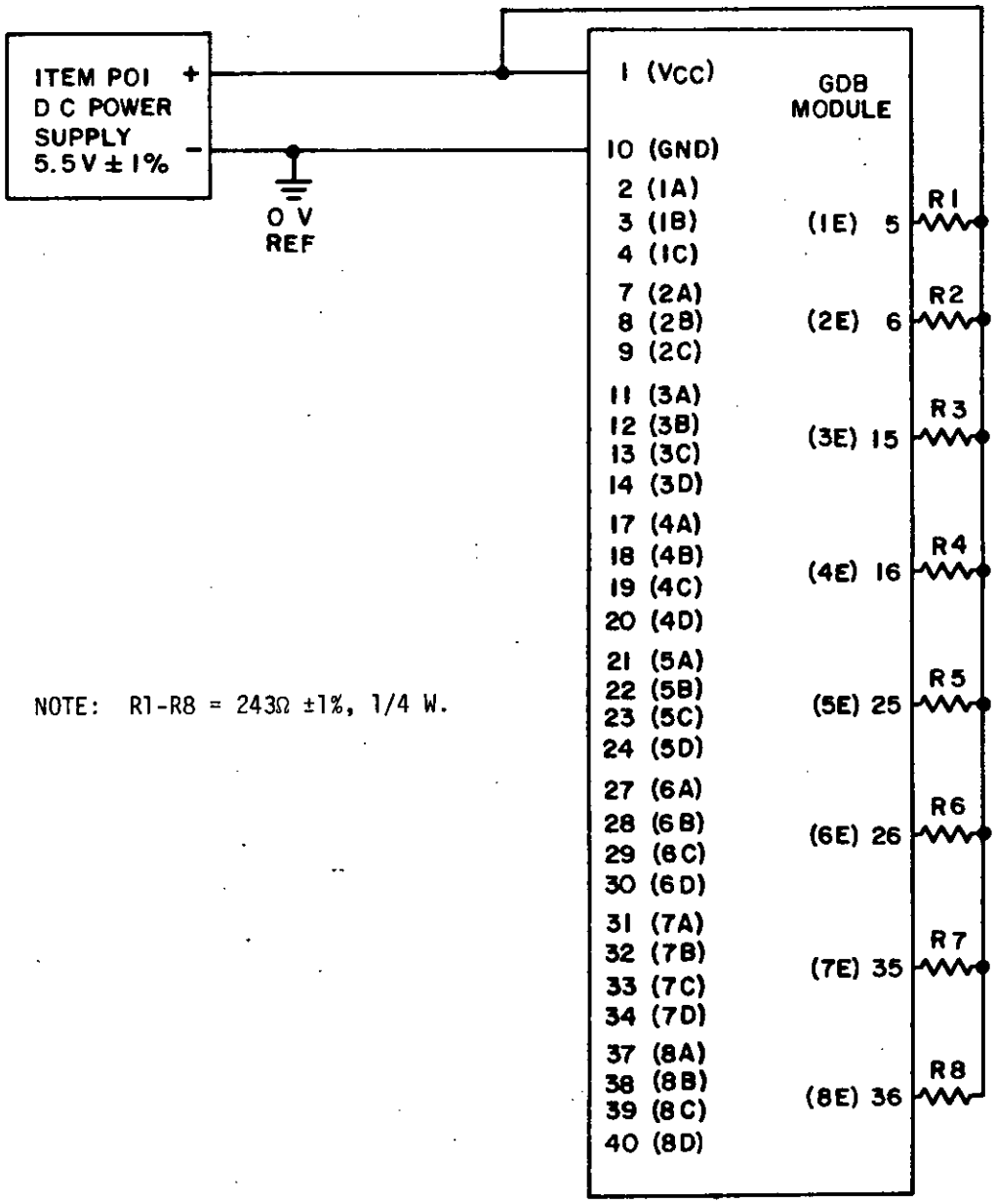
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NOTES:
1. R1-R8 = 243Ω ±1%, 1/4 W.
2. C1-C8 = 130 pF ±20%.

* FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R8 = 243Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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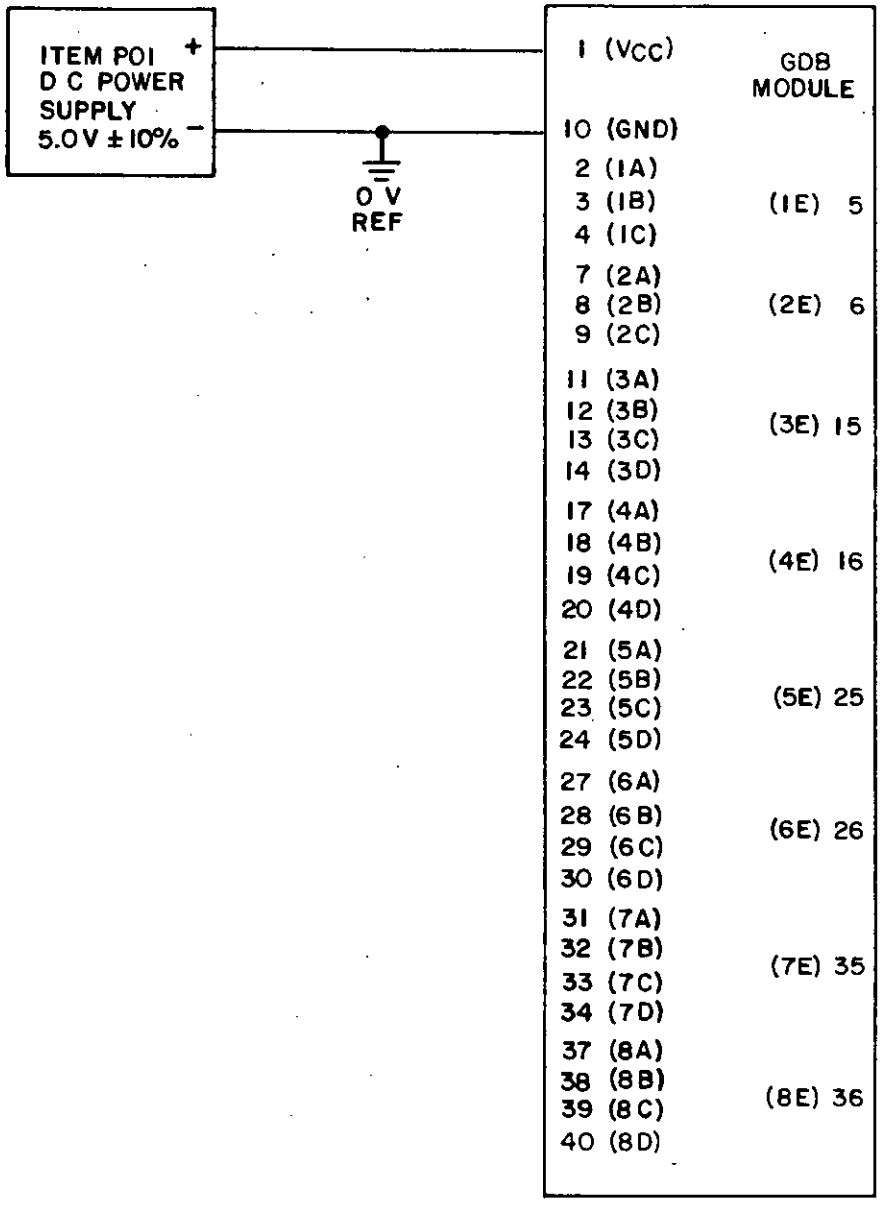


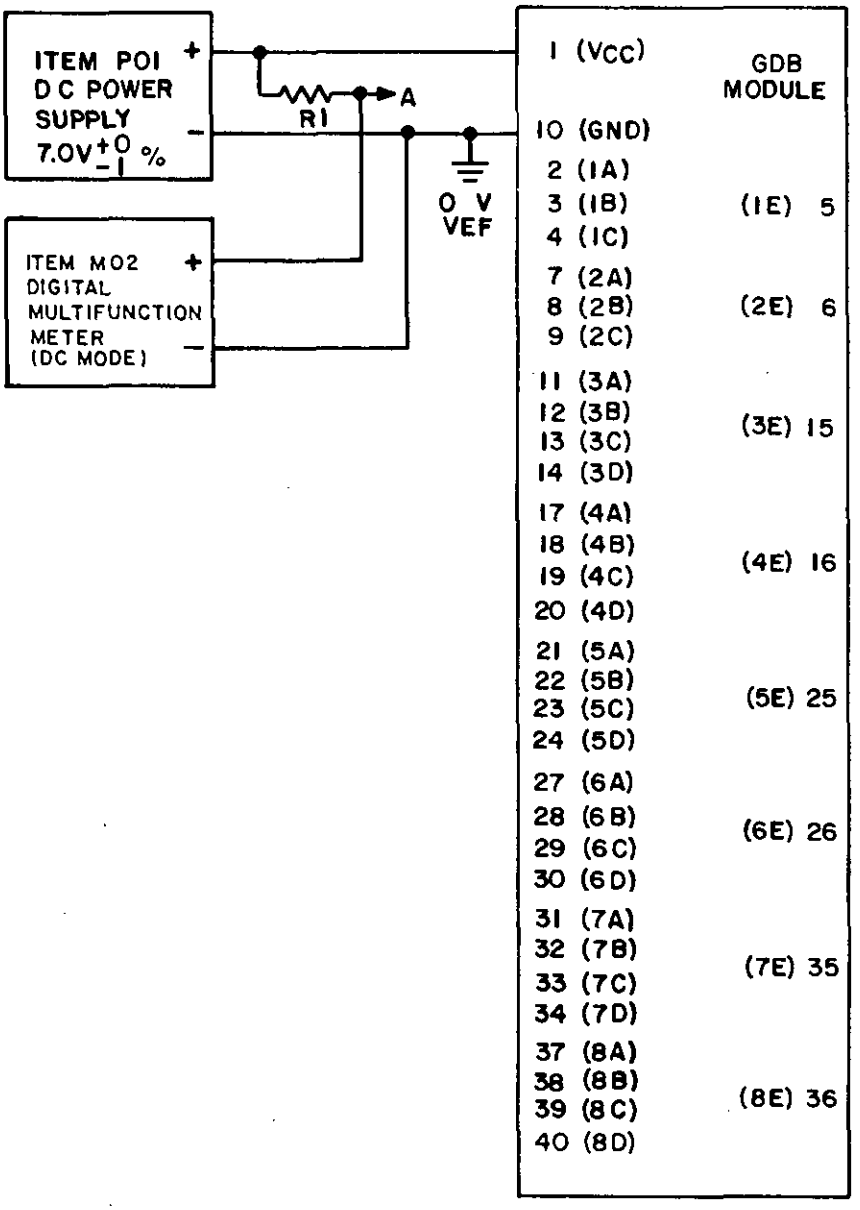
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	1E V_{OUT} (low loaded)	5	0.1	0.8	V	5
A2	2E	6				
A3	3E	15				
A4	4E	16				
A5	5E	25				
A6	6E	26				
A7	7E	35				
A8	8E	36				

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NOTES:
1. R1 = 243Ω ±1%, 1/4 W.
2. After point A has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of five seconds before the measurement is recorded.

* FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Input current	2	3,4	0.0	0.5	mA	6
B2	1B	3	2,4				
B3	1C	4	2,3				
B4	2A	7	8,9				
B5	2B	8	7,9				
B6	2C	9	7,8				
B7	3A	11	12,13,14				
B8	3B	12	11,13,14				
B9	3C	13	11,12,14				
B10	3D	14	11,12,13				
B11	4A	17	18,19,20				
B12	4B	18	17,19,20				
B13	4C	19	17,18,20				
B14	4D	20	17,18,19				
B15	5A	21	22,23,24				
B16	5B	22	21,23,24				
B17	5C	23	21,22,24				
B18	5D	24	21,22,23				

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TABLE IV. Input current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B19	6A	Input current	27	28,29,30	0.0	0.5	mA	6
B20	6B		28	27,29,30				
B21	6C		29	27,28,30				
B22	6D		30	27,28,29				
B23	7A		31	32,33,34				
B24	7B		32	31,33,34				
B25	7C		33	31,32,34				
B26	7D		34	31,32,33				
B27	8A		37	38,39,40				
B28	8B		38	37,39,40				
B29	8C		39	37,38,40				
B30	8D		40	37,38,39				

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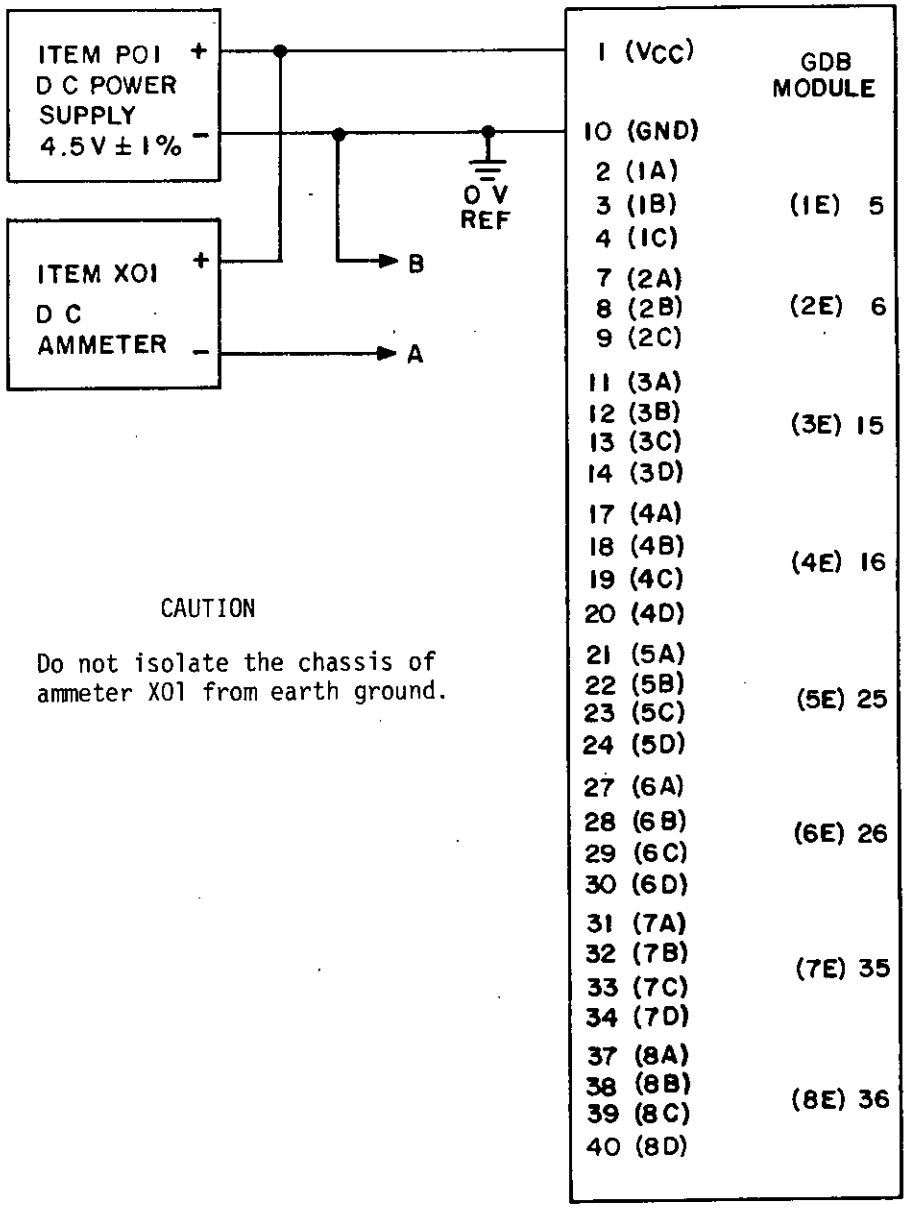


FIGURE 6. Input current test circuit for table IV.

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TABLE V. Input clamp diode voltage test.

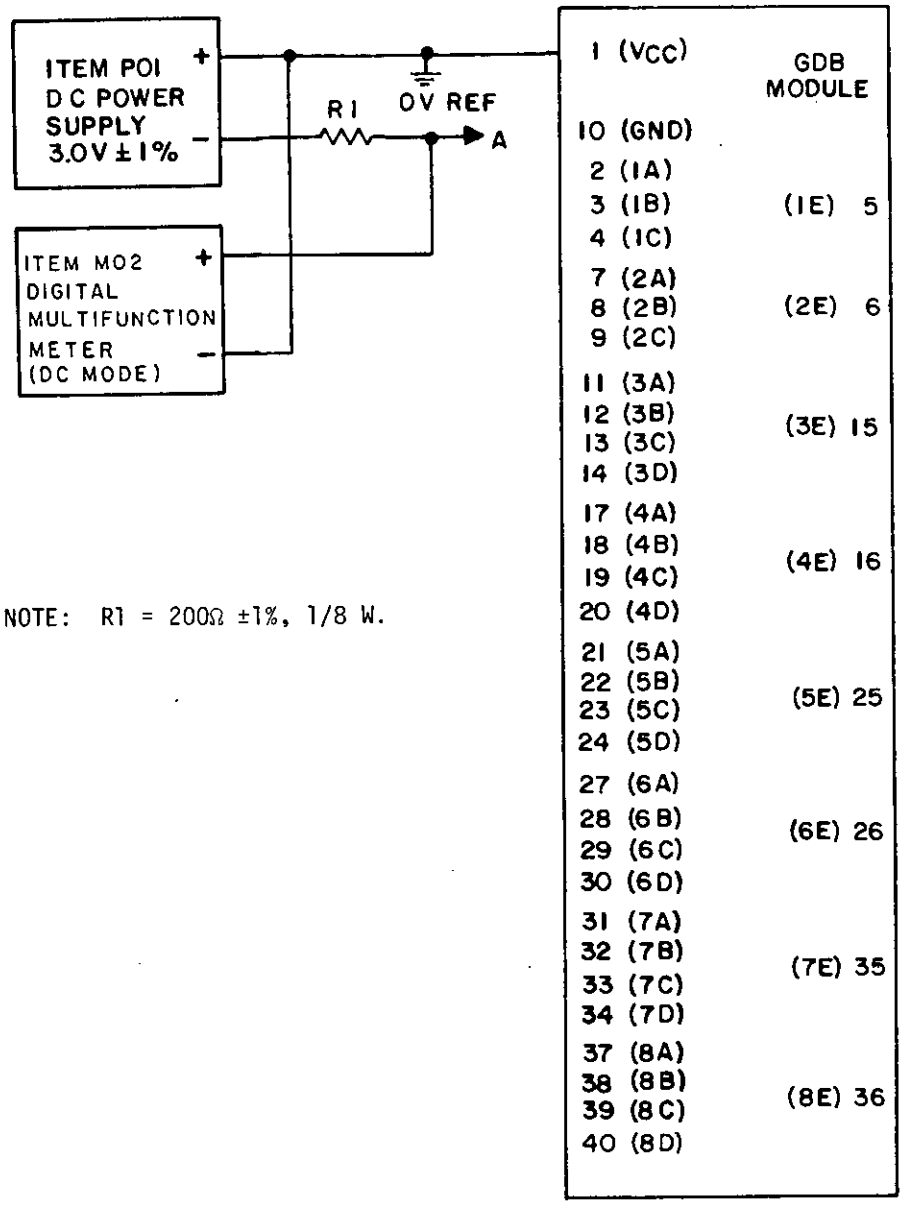
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	1A Input clamp diode voltage	2	-0.4	-1.5	V	7
C2	1B	3				
C3	1C	4				
C4	2A	7				
C5	2B	8				
C6	2C	9				
C7	3A	11				
C8	3B	12				
C9	3C	13				
C10	3D	14				
C11	4A	17				
C12	4B	18				
C13	4C	19				
C14	4D	20				
C15	5A	21				
C16	5B	22				
C17	5C	23				
C18	5D	24				

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TABLE V. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C19	6A	Input clamp diode voltage	27	-0.4	-1.5	V	7
C20	6B		28				
C21	6C		29				
C22	6D		30				
C23	7A		31				
C24	7B		32				
C25	7C		33				
C26	7D		34				
C27	8A		37				
C28	8B		38				
C29	8C		39				
C30	8D		40				

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NOTE: R1 = 200Ω ±1%, 1/8 W.

* FIGURE 7. Input clamp diode voltage test circuit for table V.

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* TABLE VI. Output voltage (low-level and (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	1E Output voltage, low level	2,3,4	5	0	450	mV	8
D2	2E	7,8,9	6				
D3	3E	11,12,13,14	15				
D4	4E	17,18,19,20	16				
D5	5E	21,22,23,24	25				
D6	6E	27,28,29,30	26				
D7	7E	31,32,33,34	35				
D8	8E	37,38,39,40	36				

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TABLE VI. Output voltage (low-level and high-level) test - Continued.

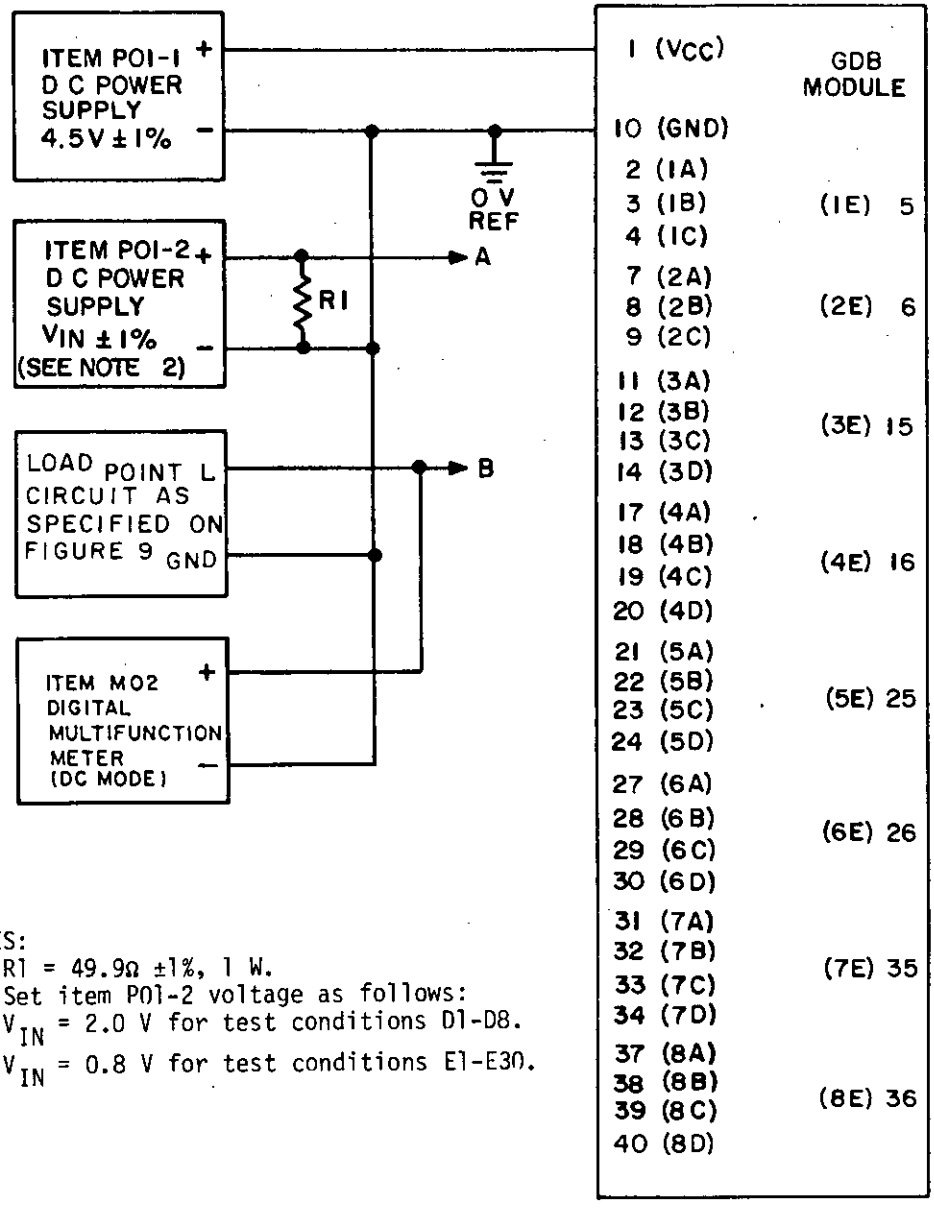
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	1E Output voltage, high level	2	5	2.4	4.5	V	8
E2	1E	3	5				
E3	1E	4	5				
E4	2E	7	6				
E5	2E	8	6				
E6	2E	9	6				
E7	3E	11	15				
E8	3E	12	15				
E9	3E	13	15				
E10	3E	14	15				
E11	4E	17	16				
E12	4E	18	16				
E13	4E	19	16				
E14	4E	20	16				
E15	5E	21	25				
E16	5E	22	25				
E17	5E	23	25				
E18	5E	24	25				
E19	6E	27	26				

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TABLE VI. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E20	6E Output voltage, high level	28	26	2.4	4.5	V	8
E21	6E	29	26				
E22	6E	30	26				
E23	7E	31	35				
E24	7E	32	35				
E25	7E	33	35				
E26	7E	34	35				
E27	8E	37	36				
E28	8E	38	36				
E29	8E	39	36				
E30	8E	40	36				

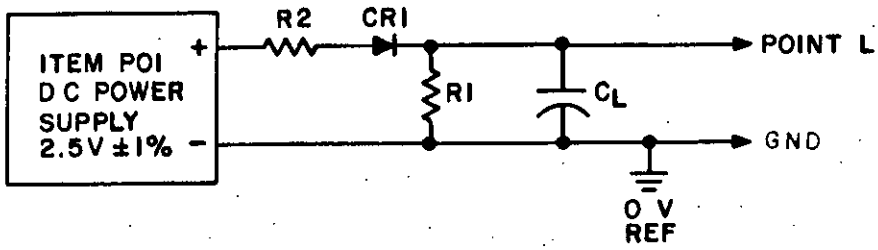
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NOTES:
1. R1 = 49.9Ω ±1%, 1 W.
2. Set item POI-2 voltage as follows:
   V_{IN} = 2.0 V for test conditions D1-D8.
   V_{IN} = 0.8 V for test conditions E1-E30.

* FIGURE 8. Output voltage (low-level and high-level) test circuit for table VI.

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NOTES:
1. CR1 = 1N4150 or equivalent.
2. C_L = 130 ±5 pF (including probe and parasitic capacitance).
3. R1 = 4.02 kΩ ±1%, 1/8 W.
4. R2 = Select value for a current flow of 22 mA ±1% from point L to ground.
5. Capacitor C_L is not required for dc electric conditions D or E.
6. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV peak.

FIGURE 9. GBD logic load circuit.

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TABLE VII. Input current (low-level and high-level) and output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1A Input current, low level	3,4	2	Open	1.0	2.2	mA	10
F2	1B	2,4	3					
F3	1C	2,3	4					
F4	2A	8,9	7					
F5	2B	7,9	8					
F6	2C	7,8	9					
F7	3A	12,13,14	11					
F8	3B	11,13,14	12					
F9	3C	11,12,14	13					
F10	3D	11,12,13	14					
F11	4A	18,19,20	17					
F12	4B	17,19,20	18					
F13	4C	17,18,20	19					
F14	4D	17,18,19	20					
F15	5A	22,23,24	21					
F16	5B	21,23,24	22					
F17	5C	21,22,24	23					
F18	5D	21,22,23	24					
F19	6A	28,29,30	27					

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F20	6B Input current, low level	27,29,30	28	Open	1.0	2.2	mA	10
F21	6C	27,28,30	29					
F22	6D	27,28,29	30					
F23	7A	32,33,34	31					
F24	7B	31,33,34	32					
F25	7C	31,32,34	33					
F26	7D	31,32,33	34					
F27	8A	38,39,40	37					
F28	8B	37,39,40	38					
F29	8C	37,38,40	39					
F30	8D	37,38,39	40					

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1A Input current, high level	2	Open	3	0	60	µA	10
G2	1A	2		4				
G3	1B	3		2				
G4	1B	3		4				
G5	1C	4		2				
G6	1C	4		3				
G7	2A	7		8				
G8	2A	7		9				
G9	2B	8		7				
G10	2B	8		9				
G11	2C	9		7				
G12	2C	9		8				
G13	3A	11		12				
G14	3A	11		13				
G15	3A	11		14				
G16	3B	12		11				
G17	3B	12		13				
G18	3B	12		14				
G19	3C	13		11				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G20	3C Input current, high level	13	Open	12	0	60	µA	10
G21	3C	13		14				
G22	3D	14		11				
G23	3D	14		12				
G24	3D	14		13				
G25	4A	17		18				
G26	4A	17		19				
G27	4A	17		20				
G28	4B	18		17				
G29	4B	18		19				
G30	4B	18		20				
G31	4C	19		17				
G32	4C	19		18				
G33	4C	19		20				
G34	4D	20		17				
G35	4D	20		18				
G36	4D	20		19				
G37	5A	21		22				
G38	5A	21		23				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G39	5A Input current, high level	21	Open	24	0	60	µA	10
G40	5B	22		21				
G41	5B	22		23				
G42	5B	22		24				
G43	5C	23		21				
G44	5C	23		22				
G45	5C	23		24				
G46	5D	24		21				
G47	5D	24		22				
G48	5D	24		23				
G49	6A	27		28				
G50	6A	27		29				
G51	6A	27		30				
G52	6B	28		27				
G53	6B	28		29				
G54	6B	28		30				
G55	6C	29		27				
G56	6C	29		28				
G57	6C	29		30				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G58	6D Input current, high level	30	Open	27	0	60	µA	10
G59	6D	30		28				
G60	6D	30		29				
G61	7A	31		32				
G62	7A	31		33				
G63	7A	31		34				
G64	7B	32		31				
G65	7B	32		33				
G66	7B	32		34				
G67	7C	33		31				
G68	7C	33		32				
G69	7C	33		34				
G70	7D	34		31				
G71	7D	34		32				
G72	7D	34		33				
G73	8A	37		38				
G74	8A	37		39				
G75	8A	37		40				
G76	8B	38		37				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

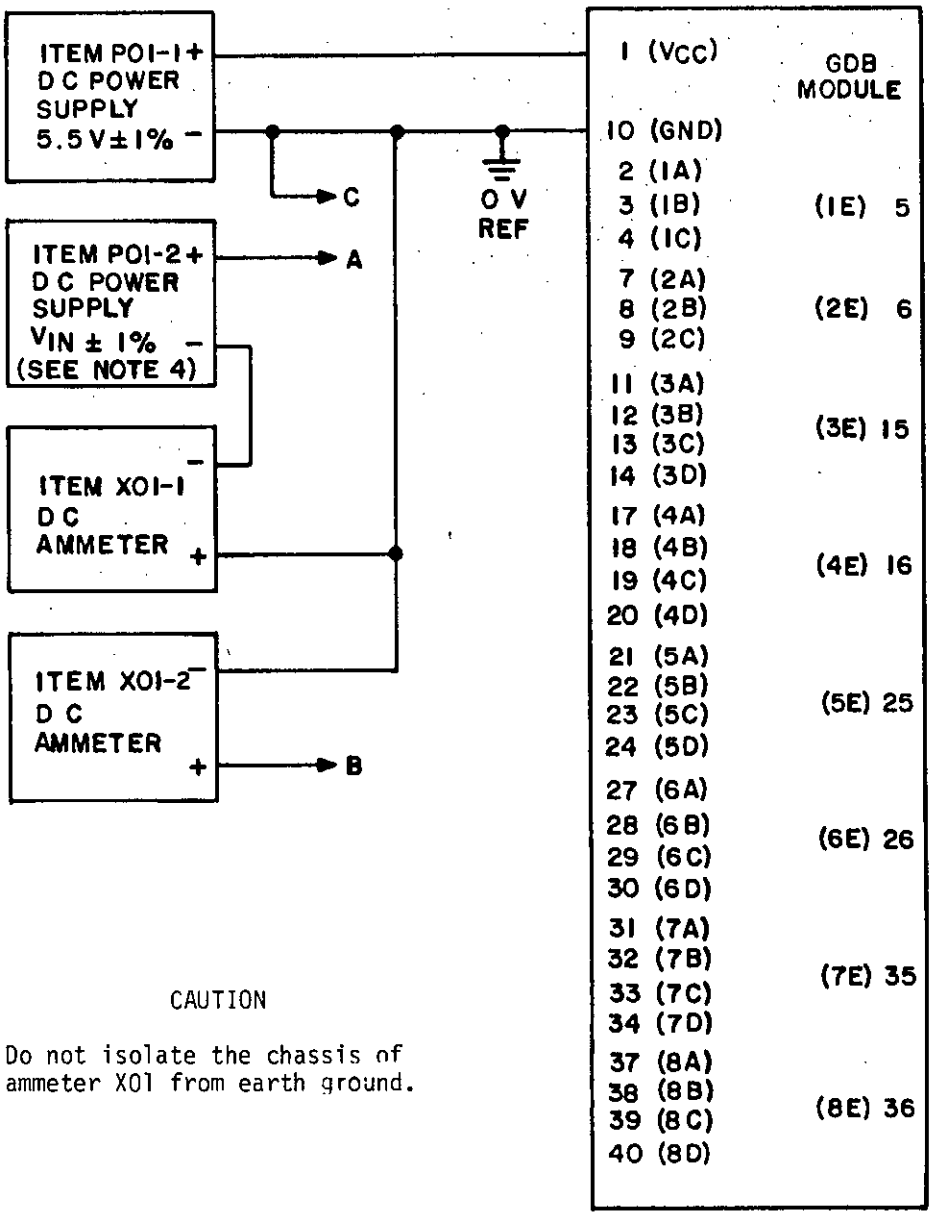
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G77	8B Input current, high level	38	Open	39	0	60	µA	10
G78	8B	38		40				
G79	8C	39		37				
G80	8C	39		38				
G81	8C	39		40				
G82	8D	40		37				
G83	8D	40		38				
G84	8D	40		39				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1E Output short circuit current	Open	5	2,3,4	40	100	mA	10
H2	2E		6	7,8,9				
H3	3E		15	11,12,13, 14				
H4	4E		16	17,18,19, 20				
H5	5E		25	21,22,23, 24				
H6	6E		26	27,28,29, 30				
H7	7E		35	31,32,33, 34				
H8	8E		36	37,38,39, 40				

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since ammeters X01-1 and X01-2 are not used simultaneously, only one ammeter X01 is required.
2. With point A disconnected, ammeter X01-1 should indicate 0.0 current on the 1.0 µA range.
3. When performing output short circuit current test, only one output shall be shorted at a time, and the measurement taken within five seconds.
4. Set item P01-2 voltage as follows:
   V_{IN} = 4.5 V for test conditions F1-F30.
   V_{IN} = 2.4 V for test conditions G1-G84.
5. Read meter item X01-2 for test conditions F1-F30 and H1-H8 and meter item X01-1 for test conditions G1-G84.

* FIGURE 10. Input current (low-level and high-level) and output short circuit current test circuit for table VII.

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TABLE VIII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
J1	Power supply current, low level	B	20	90	mA	11
J2	Power supply current, high level	10	10	44	mA	11

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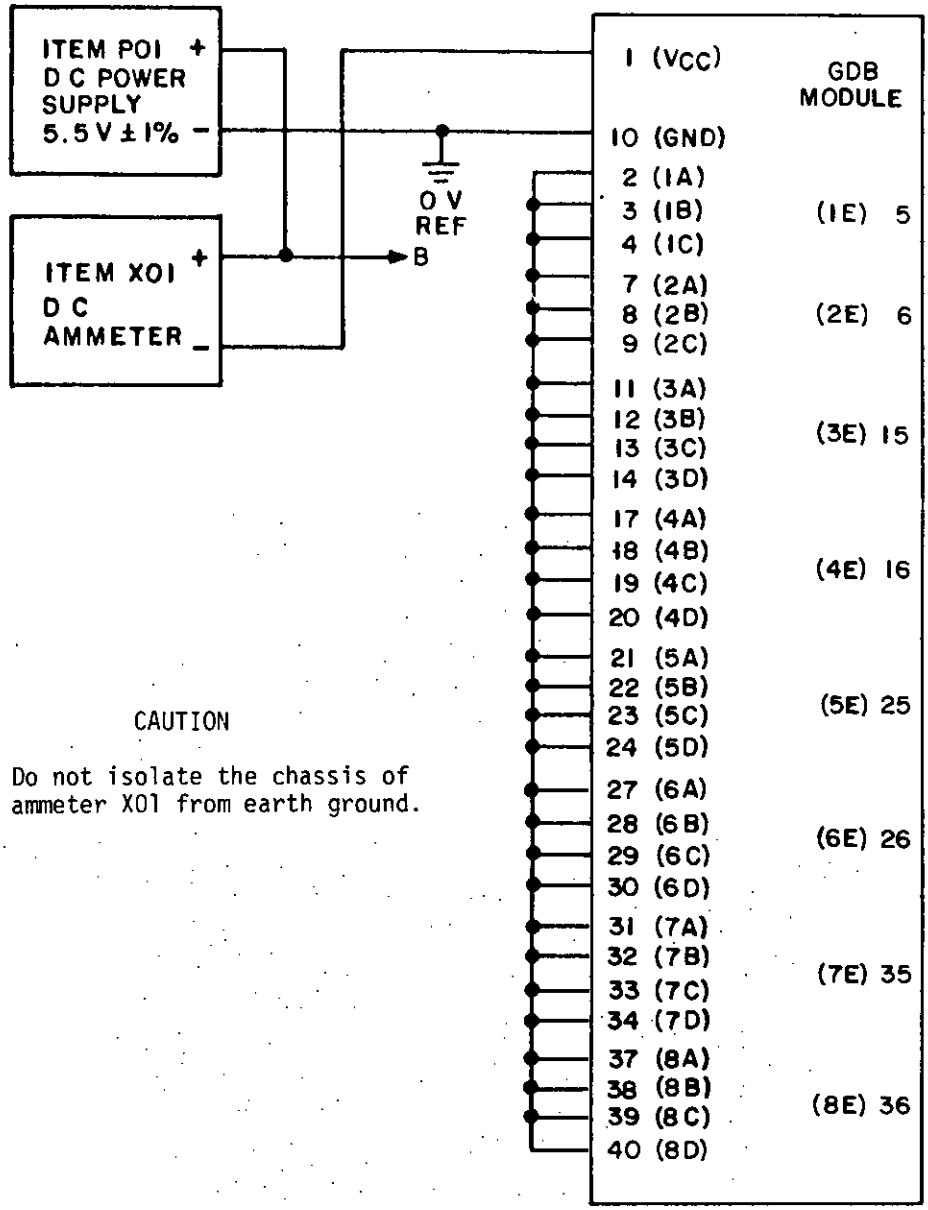


FIGURE 11. Power supply current (low-level and high-level) test circuit for table VIII.

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* TABLE IX. Transition time (t_{THL} and t_{TLH}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	1A Transition time (t_{THL})	2	5	7,11,17,21, 27,31,37	3	20	ns	12,13
K2	2A	7	6	2,11,17,21, 27,31,37				
K3	3A	11	15	2,7,17,21, 27,31,37				
K4	4A	17	16	2,7,11,21, 27,31,37				
K5	5A	21	25	2,7,11,17, 27,31,37				
K6	6A	27	26	2,7,11,17, 21,31,37				
K7	7A	31	35	2,7,11,17, 21,27,37				
K8	8A	37	36	2,7,11,17, 21,27,31				

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* TABLE IX. Transition time (t_{THL} and t_{TLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	1A Transition time (t_{TLH})	2	5	7,11,17,21, 27,31,37	3	20	ns	12,13
L2	2A	7	6	2,11,17,21, 27,31,37				
L3	3A	11	15	2,7,17,21, 27,31,37				
L4	4A	17	16	2,7,11,21, 27,31,37				
L5	5A	21	25	2,7,11,17, 27,31,37				
L6	6A	27	26	2,7,11,17, 21,31,37				
L7	7A	31	35	2,7,11,17, 21,27,37				
L8	8A	37	36	2,7,11,17, 21,27,31				

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* TABLE X. Propagation delay time (t_{PHL} and t_{PLH}) test.

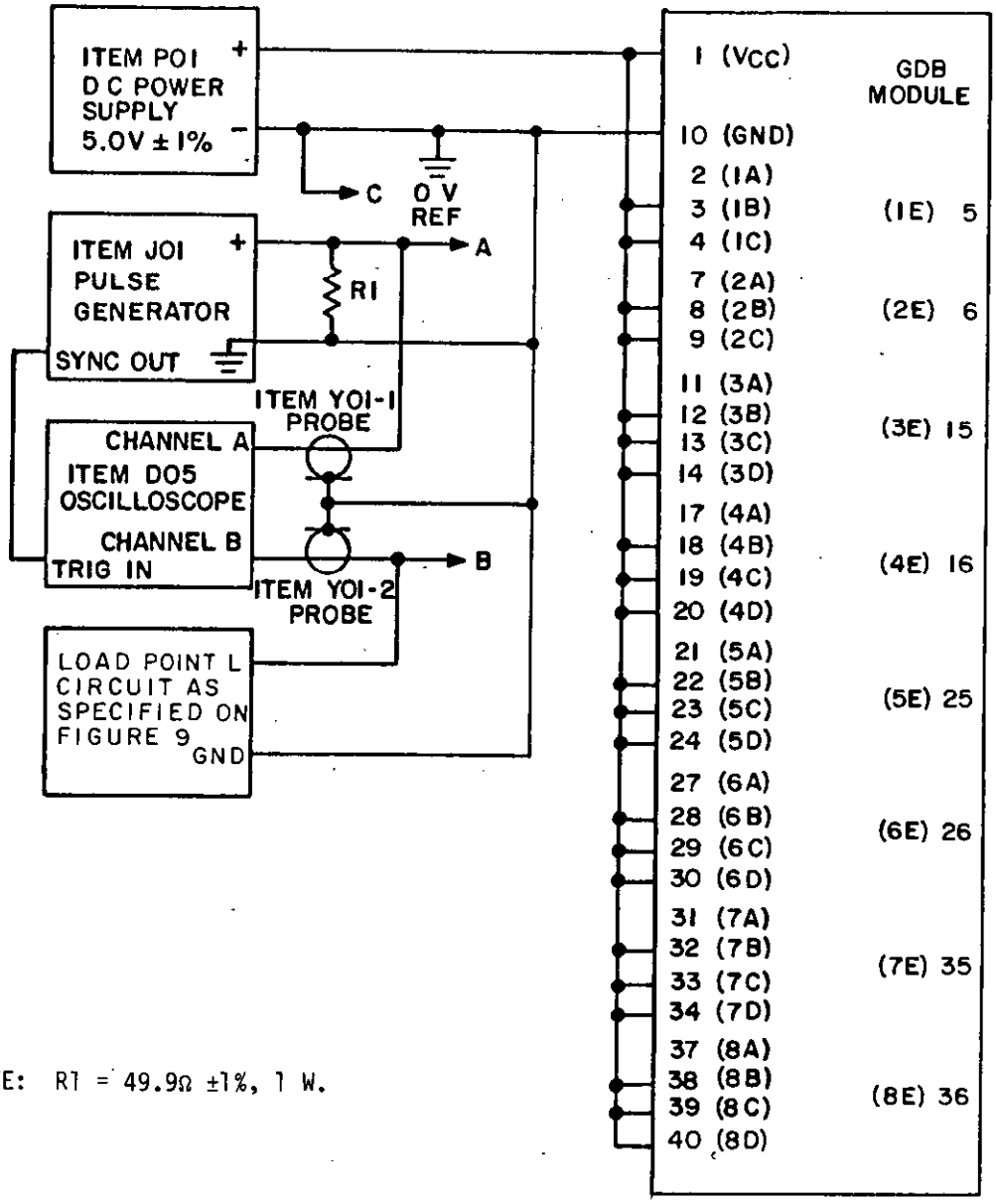
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	1A Propagation delay time (t_{PHL})	2	5	7,11,17,21, 27,31,37	3	20	ns	12,13
M2	2A	7	6	2,11,17,21, 27,31,37				
M3	3A	11	15	2,7,17,21, 27,31,37				
M4	4A	17	16	2,7,11,21, 27,31,37				
M5	5A	21	25	2,7,11,17, 27,31,37				
M6	6A	27	26	2,7,11,17, 21,31,37				
M7	7A	31	35	2,7,11,17, 21,27,37				
M8	8A	37	36	2,7,11,17, 21,27,31				

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* TABLE X. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	1A Propagation delay time (t_{PLH})	2	5	7,11,17,21, 27,31,37	3	20	ns	12,13
N2	2A	7	6	2,11,17,21, 27,31,37				
N3	3A	11	15	2,7,17,21, 27,31,37				
N4	4A	17	16	2,7,11,21, 27,31,37				
N5	5A	21	25	2,7,11,17, 27,31,37				
N6	6A	27	26	2,7,11,17, 21,31,37				
N7	7A	31	35	2,7,11,17, 21,27,37				
N8	8A	37	36	2,7,11,17, 21,27,31				

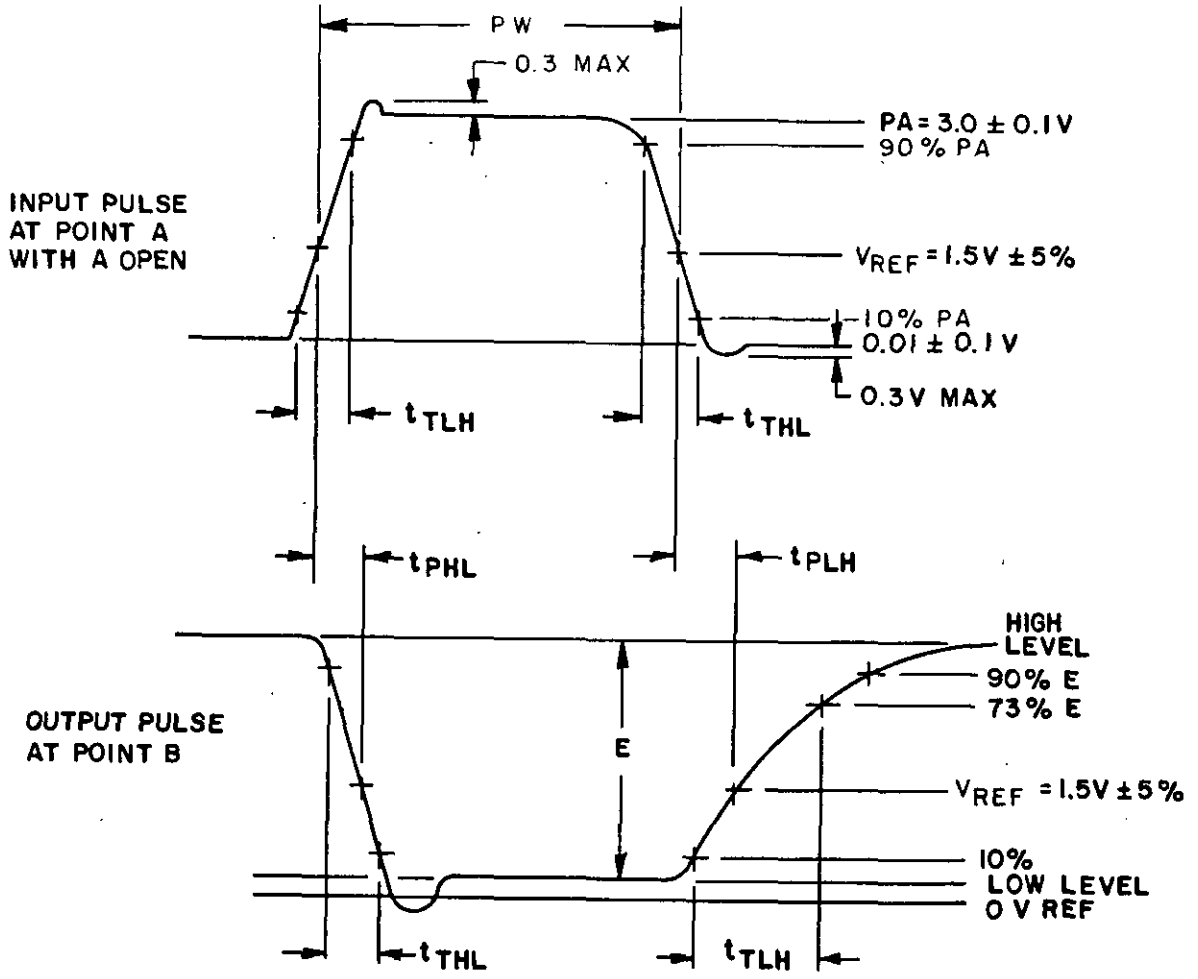
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NOTE: R1 = 49.9Ω ±1%, 1 W.

* FIGURE 12. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables IX and X.

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NOTES:
1. Input pulse parameters:
   PW = 500 ns ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PRR = 1.0 MHz ±10%.
2. Before making measurements, verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.45 V.
3. All voltage levels are referenced to ground.

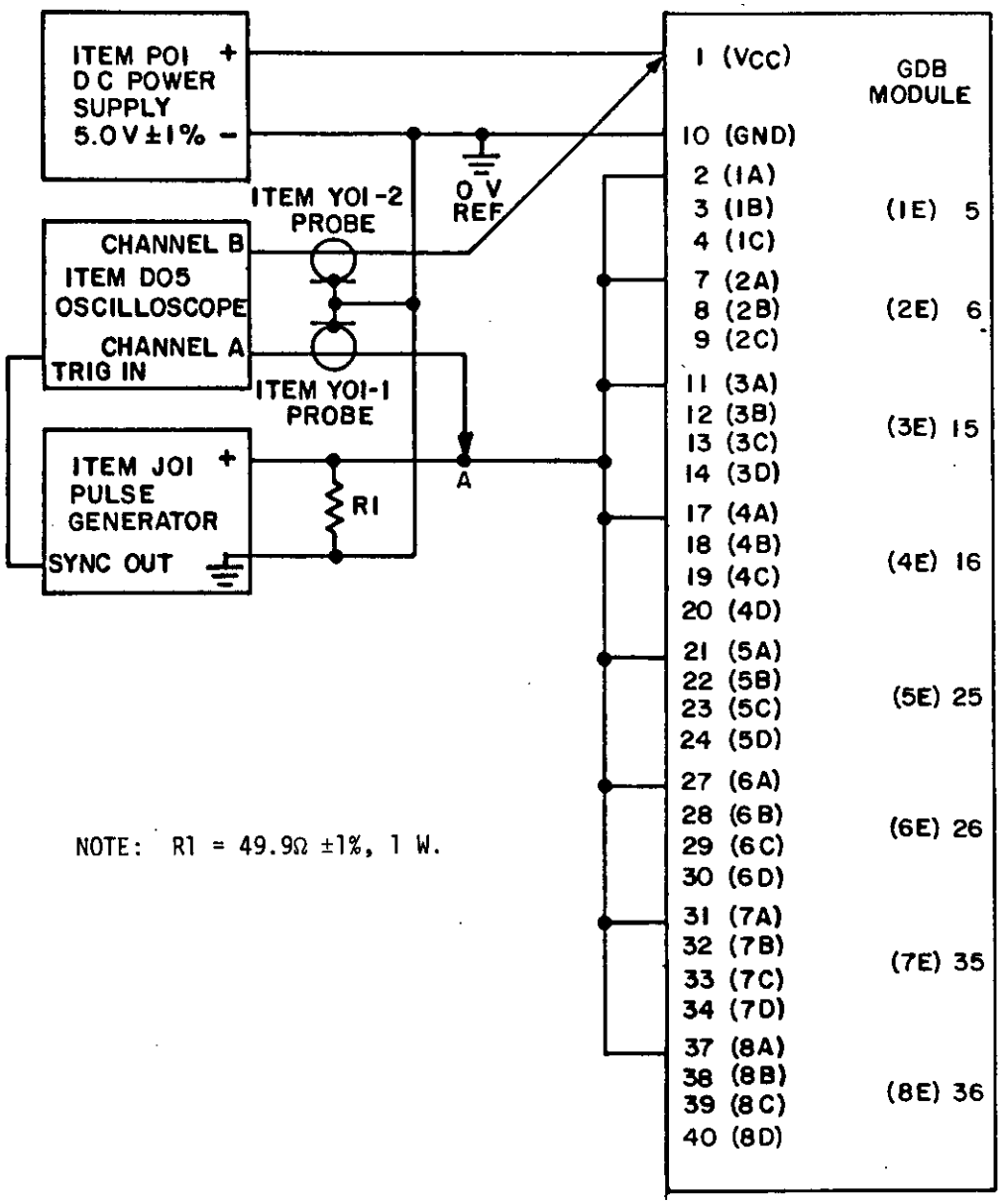
FIGURE 13. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables IX and X.

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TABLE XI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
P	Power supply transient amplitude	0	400	mV	14,15

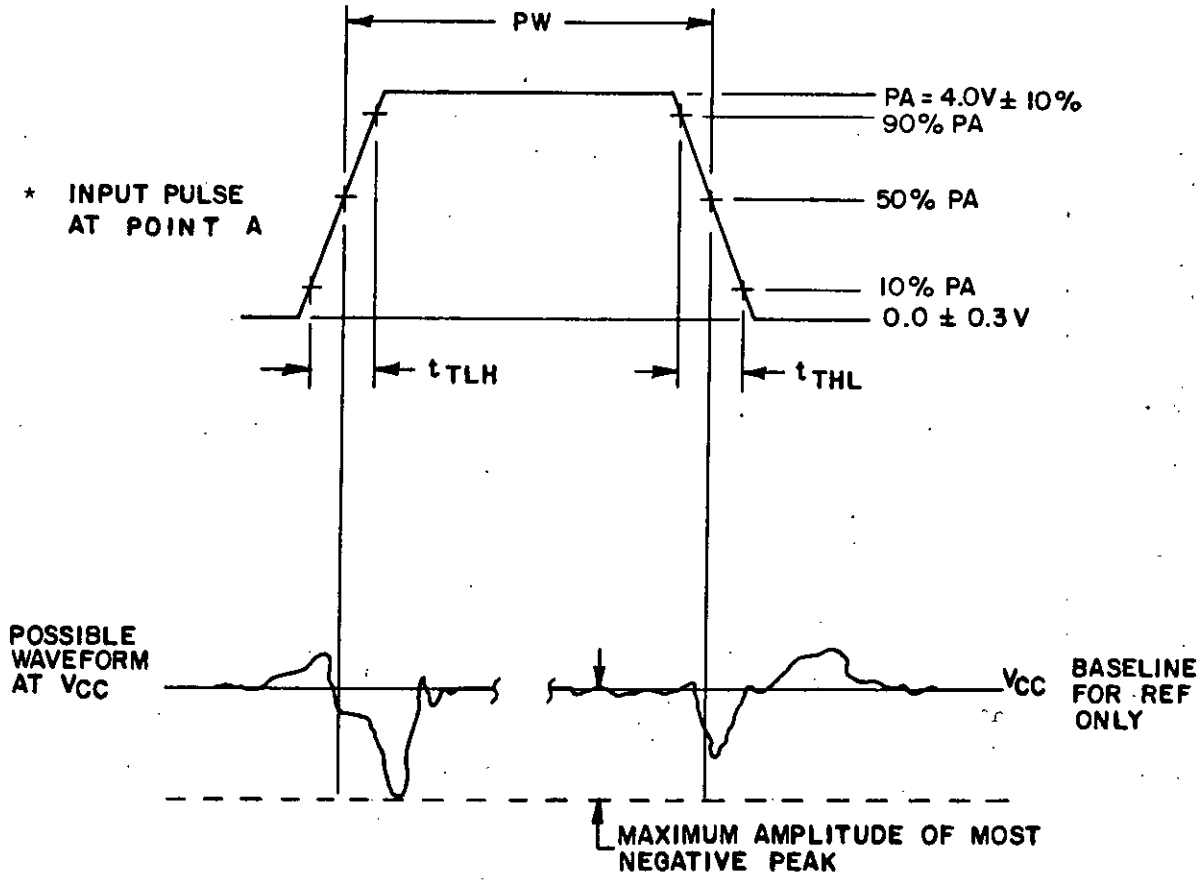
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NOTE: R1 = 49.9Ω ±1%, 1 W.

FIGURE 14. Power supply transient amplitude test circuit for table XI.

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NOTE: Input pulse parameters:
PW = 3 µs ±10%.
$t_{TLH} = t_{THL}$ = 10 ±1 ns.
PRR = 100 kHz ±10%.

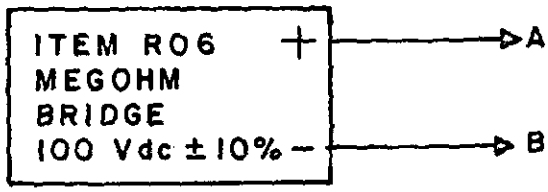
FIGURE 15. Power supply transient amplitude test circuit waveforms for table XI.

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* TABLE XII. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q	Isolation	All pins	Frame	10	∞	MΩ	16

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WARNING

When using a megohm bridge high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

	GDB MODULE
1 (Vcc)	
10 (GND)	
2 (1A)	
3 (1B)	(1E) 5
4 (1C)	
7 (2A)	
8 (2B)	(2E) 6
9 (2C)	
11 (3A)	
12 (3B)	(3E) 15
13 (3C)	
14 (3D)	
17 (4A)	
18 (4B)	(4E) 16
19 (4C)	
20 (4D)	
21 (5A)	
22 (5B)	(5E) 25
23 (5C)	
24 (5D)	
27 (6A)	
28 (6B)	(6E) 26
29 (6C)	
30 (6D)	
31 (7A)	
32 (7B)	(7E) 35
33 (7C)	
34 (7D)	
37 (8A)	
38 (8B)	(8E) 36
39 (8C)	
40 (8D)	

* FIGURE 16. Isolation test circuit for table XII.

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TABLE XIII. Output crosstalk (low-level and high-level) test.

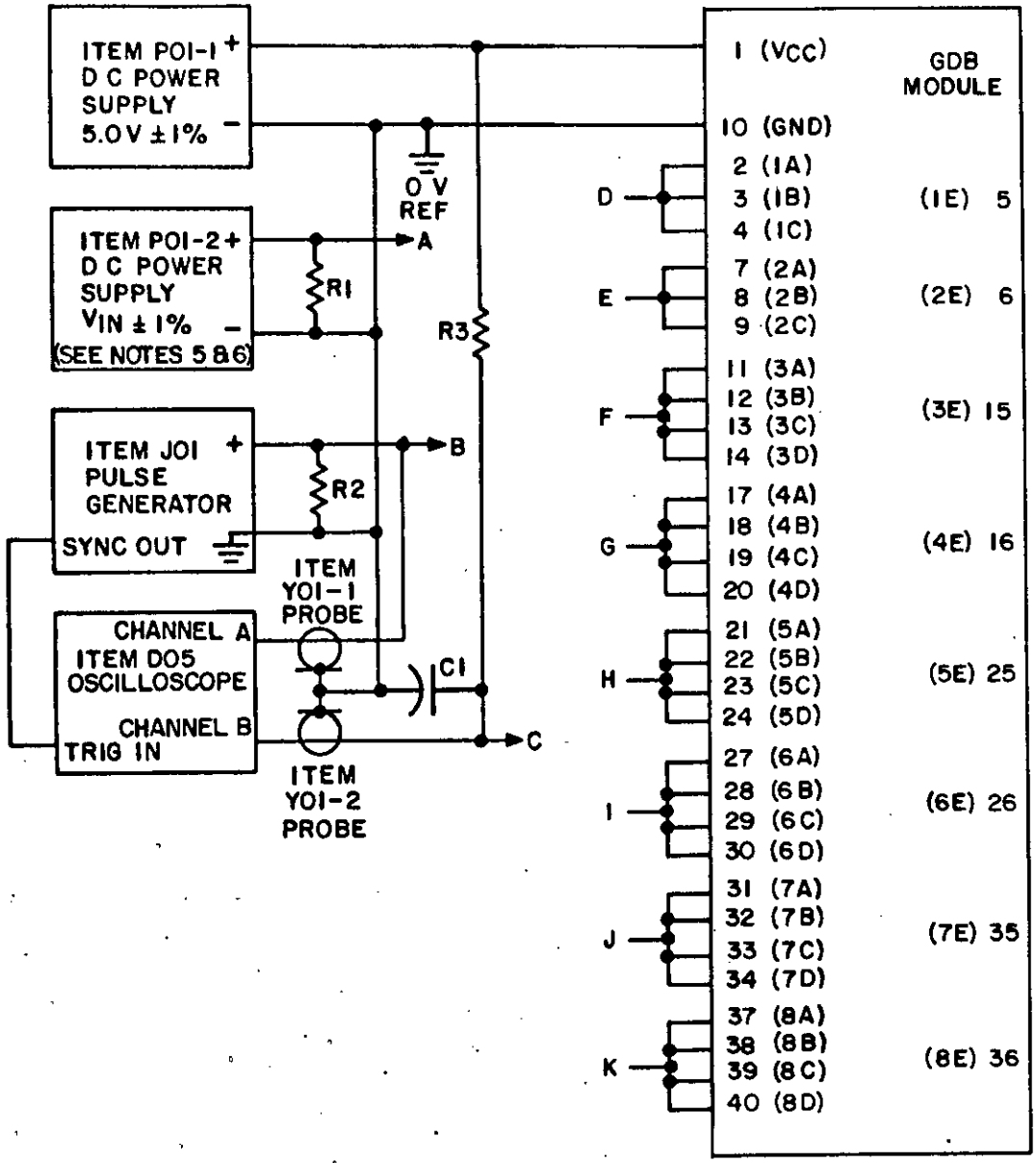
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	1E Output crosstalk, low level	D	E,F,G,H,I,J,K	5	0	400	mV	17,18
R2	2E	E	D,F,G,H,I,J,K	6				
R3	3E	F	D,E,G,H,I,J,K	15				
R4	4E	G	D,E,F,H,I,J,K	16				
R5	5E	H	D,E,F,G,I,J,K	25				
R6	6E	I	D,E,F,G,H,J,K	26				
R7	7E	J	D,E,F,G,H,I,K	35				
R8	8E	K	D,E,F,G,H,I,J	36				

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TABLE XIII. Output crosstalk (low-[redacted] and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	1E Output crosstalk, high level	D	E,F,G,H,I,J,K	5	0	400	mV	17,18
S2	2E	E	D,F,G,H,I,J,K	6				
S3	3E	F	D,E,G,H,I,J,K	15				
S4	4E	G	D,E,F,H,I,J,K	16				
S5	5E	H	D,E,F,G,I,J,K	25				
S6	6E	I	D,E,F,G,H,J,K	26				
S7	7E	J	D,E,F,G,H,I,K	35				
S8	8E	K	D,E,F,G,H,I,J	36				

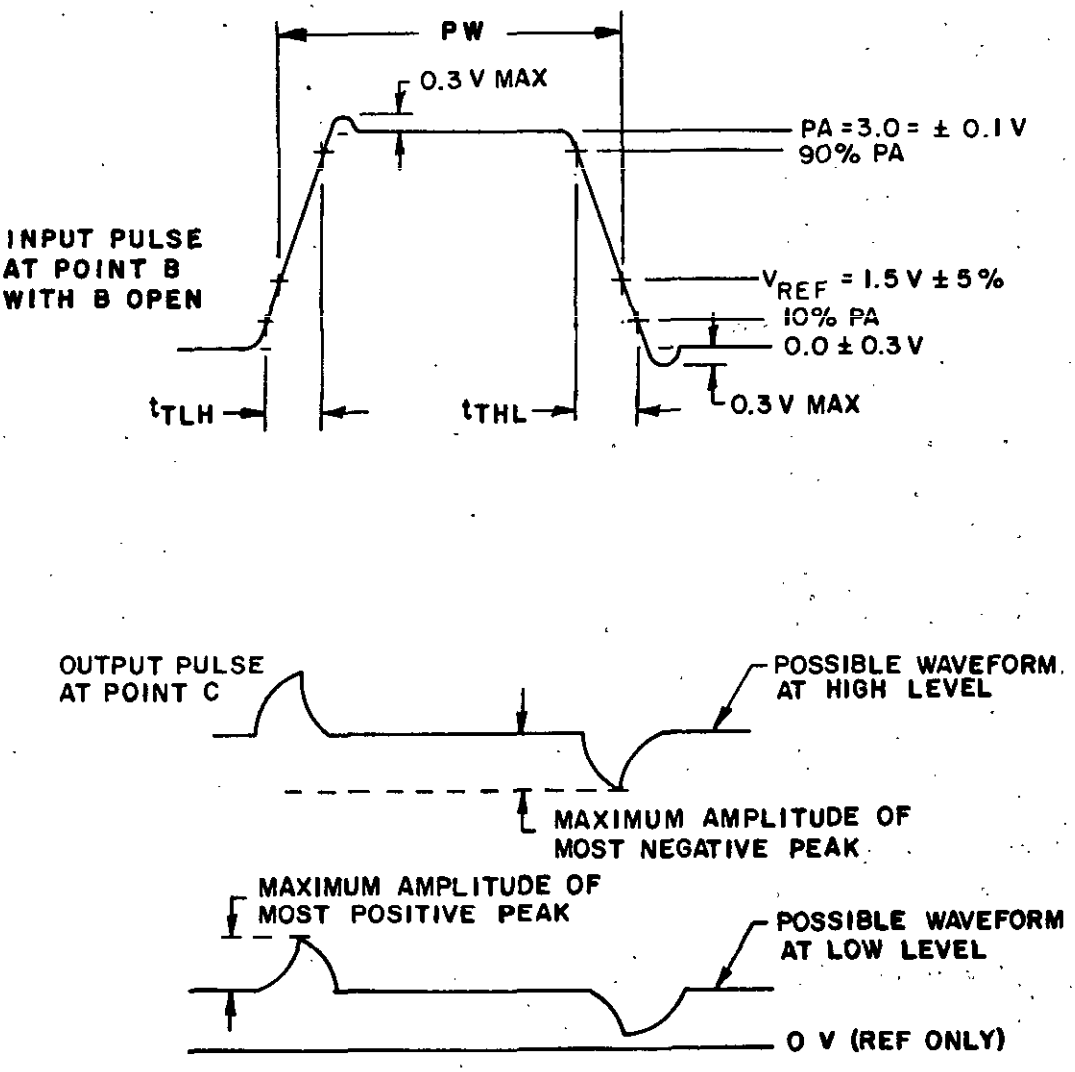
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:25Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1, R2 = 49.9Ω ±1%, 1 W.
2. R3 = 243Ω ±1%, 1/4 W.
3. C1 = 30 pF ±1% (includes probe capacitance).
4. Only the output under test shall be loaded. The loop length from pin 1 to each load resistor and output, including the length of the resistor body, shall not exceed 4.5 inches (114.3 mm).
5. Set item PO1-2 voltage as follows:
   V_{IN} = 2.4 V for test conditions R1-R8.
   V_{IN} = 0.4 V for test conditions S1-S8.

* FIGURE 17. Output crosstalk (low-level and high-level) test circuit for table XIII.

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NOTE: Input pulse parameters:
PW = 200 ns ±10%.
$t_{TLH} = t_{THL}$ = 25 ±5 ns.
PRR = 1.0 MHz ±10%.

* FIGURE 18. Output crosstalk (low-level and high-level) test circuit waveforms for table XIII.

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TABLE XIV. Input capacitance test.

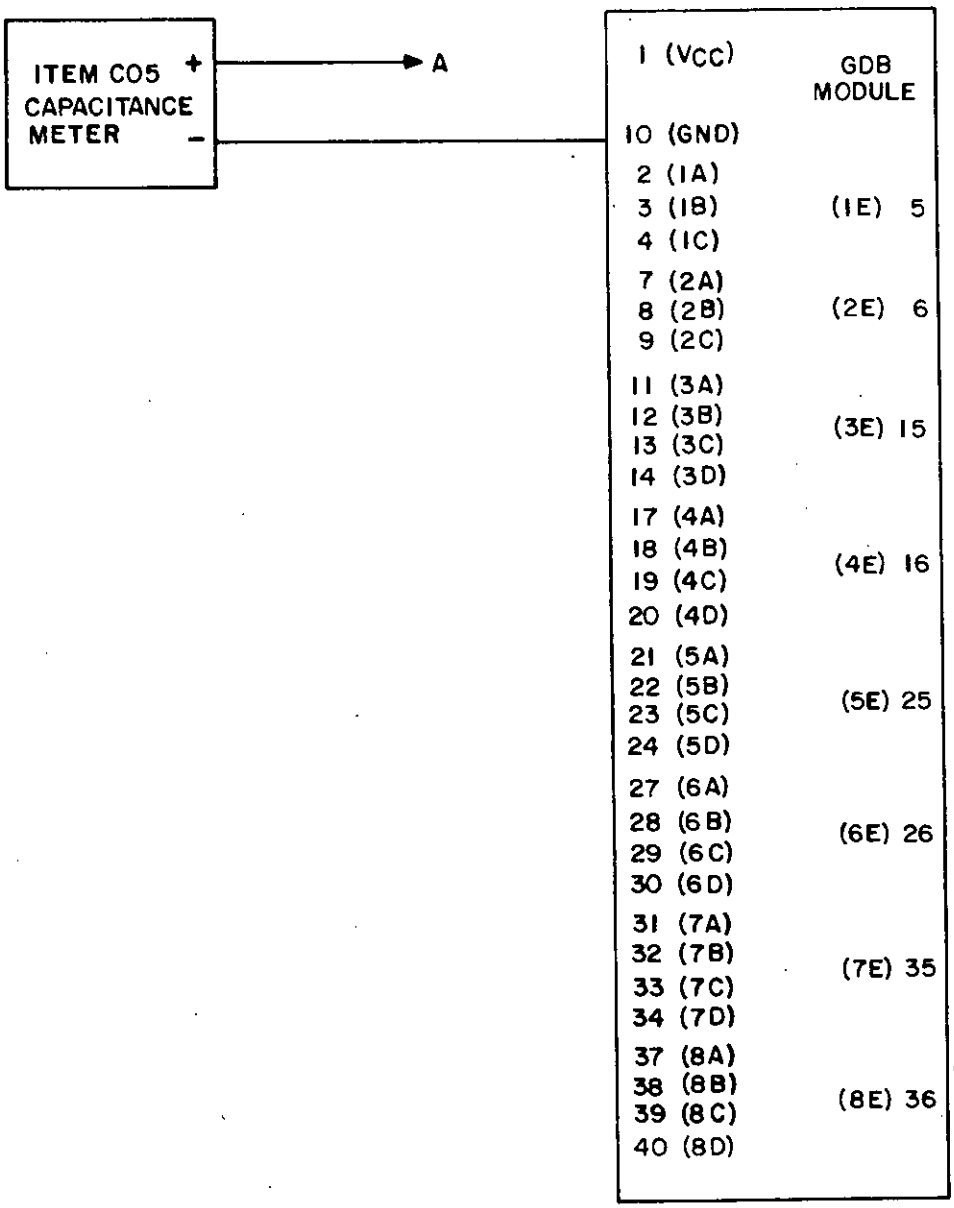
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
T1	1A Input capacitance	2	2	12	pF	19
T2	1B	3				
T3	1C	4				
T4	2A	7				
T5	2B	8				
T6	2C	9				
T7	3A	11				
T8	3B	12				
T9	3C	13				
T10	3D	14				
T11	4A	17				
T12	4B	18				
T13	4C	19				
T14	4D	20				
T15	5A	21				
T16	5B	22				
T17	5C	23				
T18	5D	24				

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TABLE XIV. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
T19	6A Input capacitance	27	2	12	pF	19
T20	6B	28				
T21	6C	29				
T22	6D	30				
T23	7A	31				
T24	7B	32				
T25	7C	33				
T26	7D	34				
T27	8A	37				
T28	8B	38				
T29	8C	39				
T30	8D	40				

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

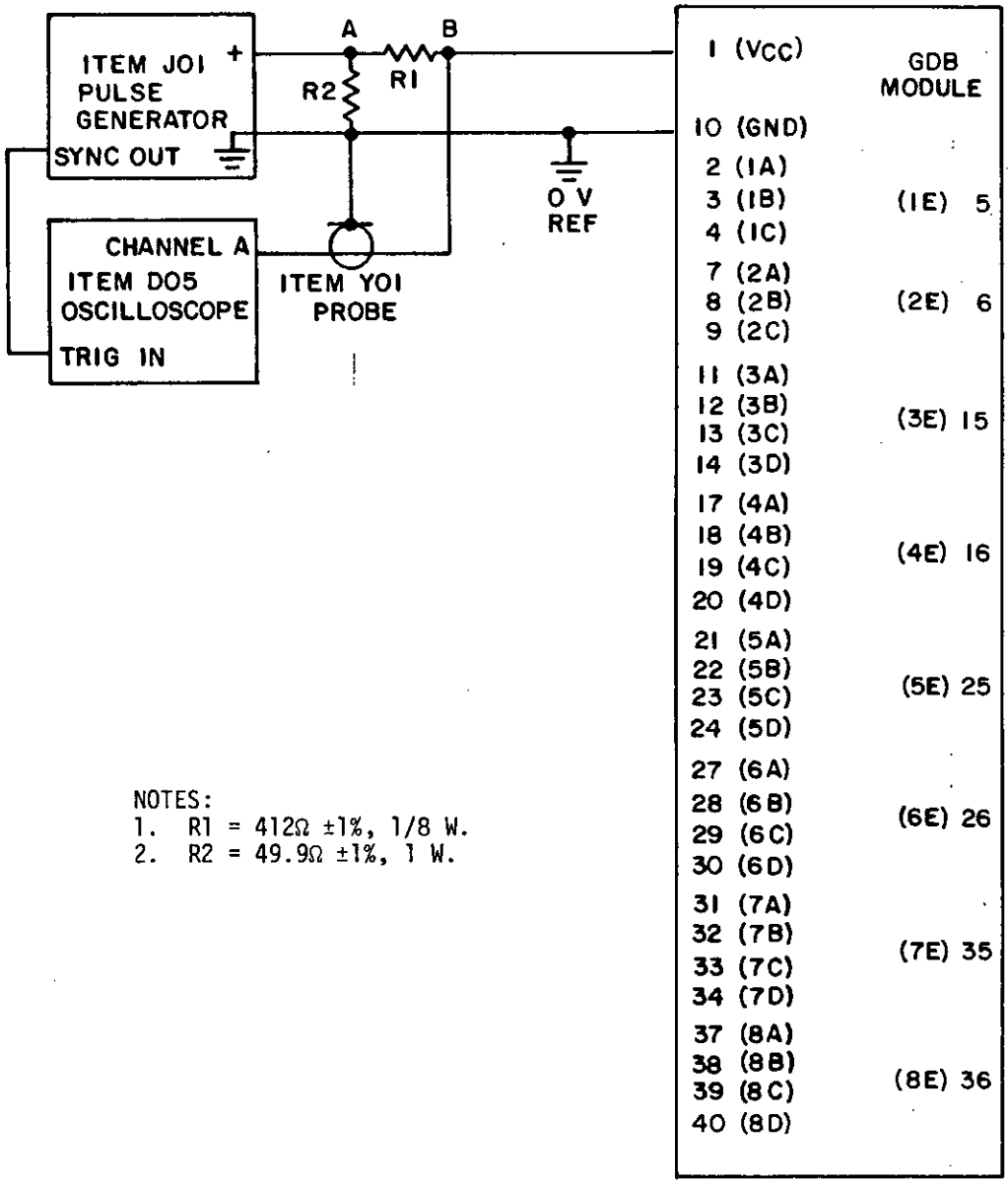
FIGURE 19. Input capacitance test circuit for table XIV.

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TABLE XV. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply transient rise time	2.5	7.5	µs	20,21

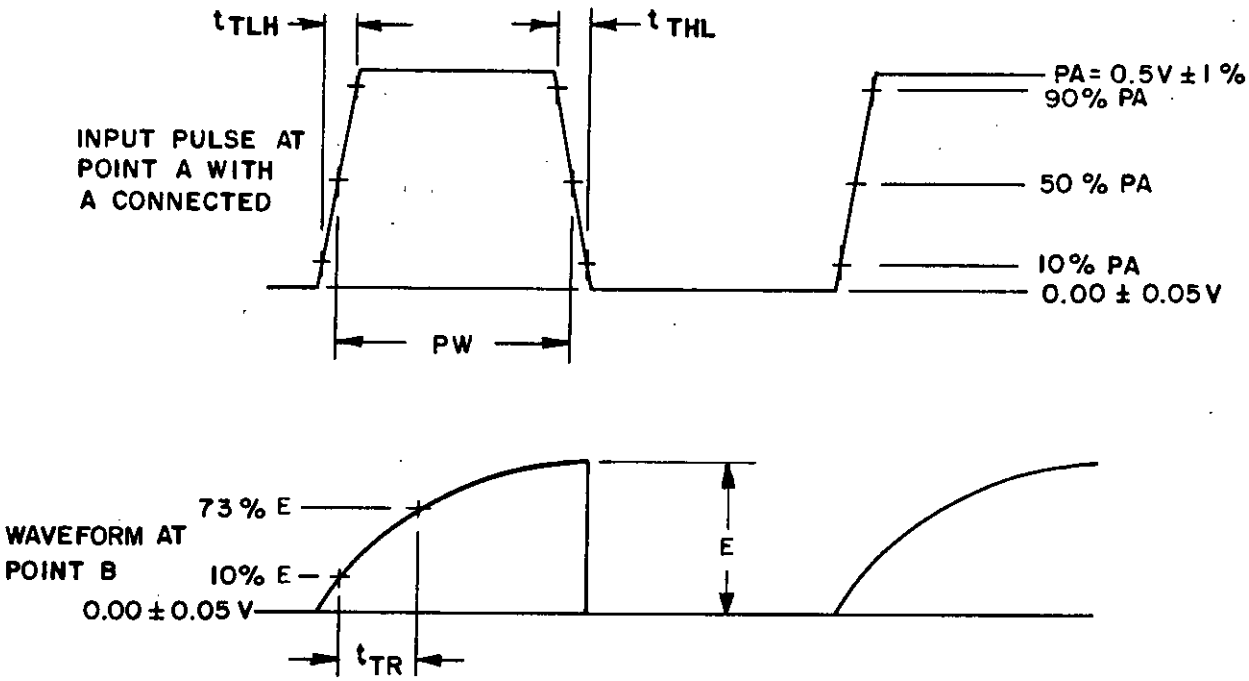
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NOTES:
1. R1 = 412Ω ±1%, 1/8 W.
2. R2 = 49.9Ω ±1%, 1 W.

FIGURE 20. Power supply transient rise time test circuit for table XV.

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NOTES:
1. Input pulse parameters:
   PW = 30 µs ±10%.
   $t_{TLH} = t_{THL}$ = 50 ns ±10%.
   PRP = 60 µs ±10%.
2. Measure transient rise time (t_{TR}) at point B.

* FIGURE 21. Power supply transient rise time test circuit waveforms for table XV.

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